CAN THE SEA HOLD MORE CO_2?

The ocean takes in 29% of the CO_2 we emit, and that uptake has stopped growing. Ocean alkalinity enhancement (OAE) might safely raise it. Nobody knows yet. This brief asks for the independent research that would find out.

29%
of the CO_2 people emitted over the past decade was taken up by the ocean

2016
the year that uptake stopped growing, at about 3.4 billion tonnes of carbon a year

10,000+
years carbon can stay dissolved as bicarbonate, if the added alkalinity persists

WHAT IT IS

OAE adds alkaline material to surface seawater, either ground minerals such as olivine or a hydroxide made with electricity. The water then holds more carbon as stable bicarbonate and draws more CO_2 from the air. It speeds up a natural process. It is not ready to deploy.

WHAT THE EVIDENCE SHOWS

- **One permitted sea trial found no harm so far.** In the Gulf of Maine in August 2025, researchers found no significant difference in plankton, fish larvae or lobster larvae. The results are preliminary and not yet peer reviewed.
- **Much was not measured:** fish, marine mammals and fisheries.
- **A lab result fell short.** A 2022 study found olivine removed about five times less carbon than theory predicts.
- **Verification is unsolved.** CarbonPlan puts confidence in verifying the electrochemical route at 3 out of 5. Project Drawdown lists OAE under Keep Watching.

WHO PAYS FOR THE RESEARCH

- **Philanthropy.** The Carbon to Sea Initiative, launched in 2023 with $50 million from foundations, is the largest funder. It also advocates for the field, and it funded the Gulf of Maine trial with ICONIQ Ocean Co-Lab.
- **Credit buyers.** In 2025 companies including Stripe, Google and Shopify agreed to pay $31.3 million for 115,211 tons of removal from one OAE company.
- **Public money** is small by comparison. We found no oil company among the named funders.

That is why the ask is for public money and independent review.

FIVE DECISIONS FOR COP31

1. **Fund independent verification research.** An open, global standard for monitoring, reporting and verification, publicly funded and independently reviewed, in place before any credit counts toward a climate target.
2. **Set ecological guardrails.** Thresholds for pH, trace metals and biodiversity that pause a trial automatically, with independent oversight.
3. **Clarify governance.** Jurisdiction, liability and consent for ocean trials under UNCLOS and the London Protocol.
4. **Share decisions and benefits.** Coastal and Indigenous communities hold real decision power over where and how trials run.
5. **Keep emission cuts first.** OAE adds to rapid decarbonization and can never replace it.

TWO MINUTES WITH YOUR DELEGATION? ASK

- Does our climate research budget fund independent work on verifying OAE, and who reviews it?
- Will we keep OAE credits out of national targets until an open, public standard exists?
- Who in our delegation follows ocean carbon removal, and will they raise research governance before COP32?

Scan for every source, the full case and a letter you can send to your country's climate focal point.

alkalinityask.life

Sources: Global Carbon Budget 2025 · National Academies, 2022 · Woods Hole Oceanographic Institution, February 2026 · MIT Technology Review, 2022 and 2023 · CarbonPlan CDR Verification Framework · Project Drawdown, 2025 · Frontier, 2025 · Carbon to Sea Initiative. Full references are on the page.

An independent brief, not affiliated with the organisations it cites. Figures as of 11 November 2026.